

Fig. 1: Circuit diagram of variable timer with music

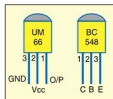


Fig. 2: Pin configurations of UM66 and BC548

IC2 is then incremented on each clock. Outputs Q1 through Q9 of IC2 go high one-by-one at interval of 30 seconds.

The rotary switch S3 selects one of these outputs and provides 3.1V power supply to melody generator IC UM66 (IC3). UM66 produces a musical signal on its output pin 1. The musical signal is amplified by transistor T1 and drives the speaker to produce the musical note.

takes Q0 to logic high. Counting starts when switch S2 is released. Output Q1 goes high on the next positive-going pulse from IC1.

When rotary switch S3 is connected to Q1 output of IC2 and reset switch S2 is pressed, a musical note is heard after a few seconds for 30 seconds. For positions 2 to 9, the silent period increases by 30 seconds at each step. At position 10 of rotary switch S3, the delay will be of four and a half minutes, followed by 30 seconds of music.

The best way to think about this is that positions 1 to 9 add 30-second delay each. If you let the timer continue to run, it repeats the tune every 5 minutes, irrespective of the setting of S3.

The circuit runs on 6V, provided by four AA cells in a battery holder. A PP3-type clip is used to connect this to the circuit. S1 is a simple single-pole,

single-throw (SPST) switch.

Assemble the circuit on a general-purpose PCB and enclose it in a suitable box. If you have a multimeter with a frequency-measuring facility, it would be ideal for setting the frequency of IC1 at pin 3 at around 0.033Hz.

Check that IC2 outputs run through the correct sequence (each high for 30 seconds). Wire the loudspeaker to its terminal pins using external wires. Fix rotary switch S3 in the front panel for selecting the time. The positions of rotary switch S3 can be labeled with stick-on tape labels. A simpler technique is to punch small discs from insulating tapes of various colours and stick these around the knob. ●

BATTERY DEEP DISCHARGE SAVER

■ T.A. BABU

This circuit can be used to ensure that a 12V sealed lead-acid battery does not go into the condition of deep discharge. The circuit disconnects the battery from load when its voltage goes below a preset

voltage (say 10.5V) level. The load remains disconnected until the battery is recharged.

Initially, when the circuit is switched on through switch S1, capacitor C1 momentarily acts as a short circuit. It connects the coil of relay RL1 to the battery. Relay RL1 energises and



connects the load to the battery. At the same time battery terminal voltage is divided by resistor R2 and the